Adversarial Search in Isolation: Minimax vs. Alpha-Beta Pruning

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Abstract

This project implements the game *Isolation* and compares two adversarial search algorithms: Minimax and Alpha-Beta pruning. We evaluate them under controlled conditions (same depth), under realistic constraints (same time budget via iterative deepening), and in gameplay against a random baseline. We also demonstrate a critical practical insight: Alpha-Beta pruning effectiveness depends strongly on move ordering. Results show that Alpha-Beta expands far fewer nodes than Minimax at the same depth, and with good move ordering can reduce expansions dramatically.

1 Introduction

Adversarial search is a core technique for decision-making in deterministic, turn-based, perfect-information games. Minimax provides an optimal decision rule (under a depth limit and heuristic evaluation). Alpha-Beta pruning is an optimization that returns the same move as Minimax at a fixed depth, but can avoid exploring large parts of the game tree.

This report covers:

- the Isolation game, including state representation, actions, and transition function,
- Minimax and Alpha-Beta in simple terms,
- the evaluation function used,
- experimental results: time, nodes explored, and depth reached,
- reflections on observed behavior and efficiency.

2 Game: Isolation

2.1 State representation

A state s stores:

- a board grid where each cell is empty, blocked, or occupied by A or B,
- the current position of each player (or `None` before initial placement),
- the active player (whose turn it is).

2.2 Initial state

The initial state is an empty board (no blocks) with both player positions unset. The first time each player moves, they **place** their piece on any empty cell.

2.3 Actions (legal moves)

Let the active player be p with current position (r, c).

- If p has not been placed yet, the action set is all empty cells.
- Otherwise, an action is a **queen move**: slide any number of squares in one of 8 directions (horizontal/vertical/diagonal) until blocked/occupied.

2.4 Transition function

Given a state s and action a (a destination cell), the transition $T(s, a) = s'$ is:

- move the active player to a,
- mark the previous player cell as blocked (unusable),
- switch the active player.

2.5 Terminal states

A state is terminal if the active player has **no legal actions**. The player to move loses; equivalently, the winner is the opponent of the active player in the terminal state.

3 Algorithms

3.1 Minimax (depth-limited)

Minimax searches a game tree to depth d, alternating:

- **max** levels: choose the move that maximizes utility for the root player,
- **min** levels: choose the move that minimizes utility (opponent response).

At depth 0 or terminal states, a heuristic evaluation estimates utility.

3.2 Alpha-Beta pruning

Alpha-Beta pruning returns the same move/value as Minimax at a fixed depth, but maintains:

- α: best value found so far for the maximizing player,
- β: best value found so far for the minimizing player.

If $\alpha \geq \beta$, the remaining sibling branches cannot affect the final choice and are pruned.

3.3 Evaluation function

We use a standard Isolation heuristic: **mobility difference**

$$h(s) = \#\text{legal moves for root} - \#\text{legal moves for opponent}.$$

This is suitable for Isolation because the game is fundamentally about restricting the opponent's mobility. Terminal states return large-magnitude utilities so that forced wins/losses dominate heuristic estimates.

3.4 Iterative deepening and "depth reached"

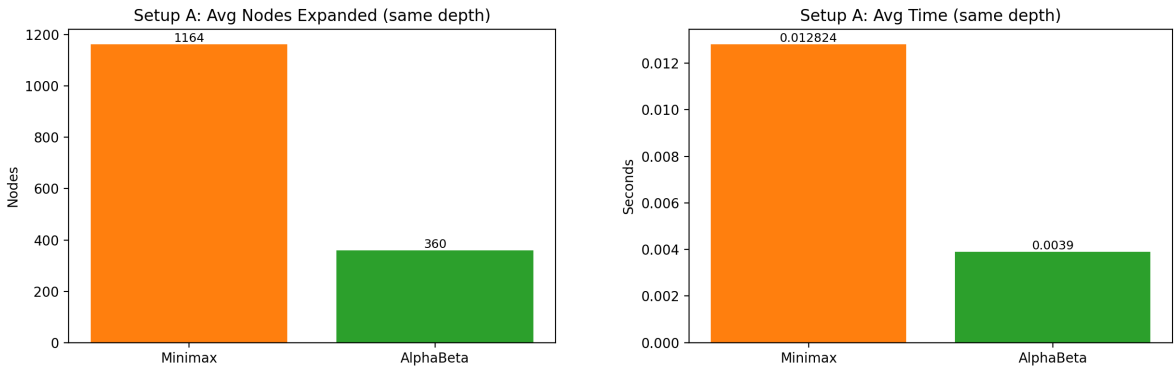
Under a per-move time budget, the agents use iterative deepening: search depth $1, 2, \ldots$ until time runs out. We record **depth reached** as the deepest fully-completed depth within the budget for that move.

4 Experimental Methodology

- **Setup A (same depth):** compare Minimax vs Alpha-Beta fairly at identical depth on identical states.
- **Setup B (same time):** compare behavior under a fixed time budget and record depth reached.
- **Setup C (vs random):** measure win rate and efficiency in gameplay against a random baseline.
- **Setup D (move ordering):** isolate the impact of move ordering on Alpha-Beta pruning.

5 Results

5.1 Setup A (same depth): nodes and time



(a) Average nodes expanded (same depth)

(b) Average runtime (same depth)

Figure 1: Setup A: Alpha-Beta expands fewer nodes and runs faster than Minimax at the same depth.

5.2 Setup B (same time): depth reached

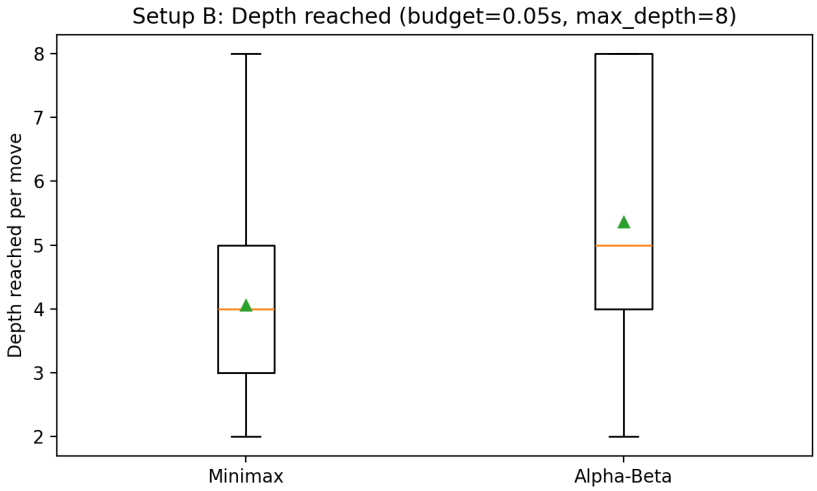
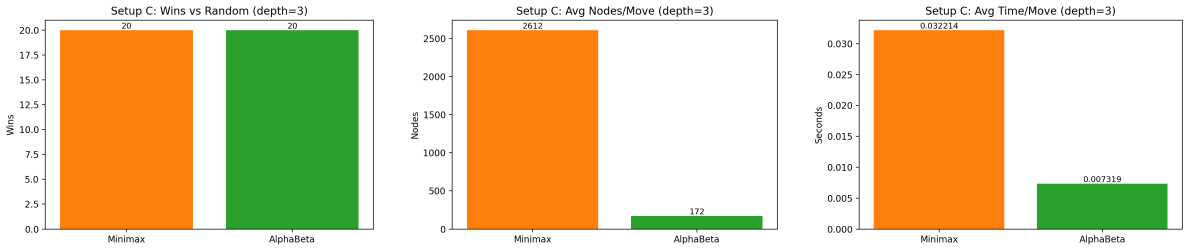


Figure 2: Setup B: Depth reached per move under the same time budget using iterative deepening (mean shown).

This directly quantifies how pruning translates into deeper search under a time constraint.

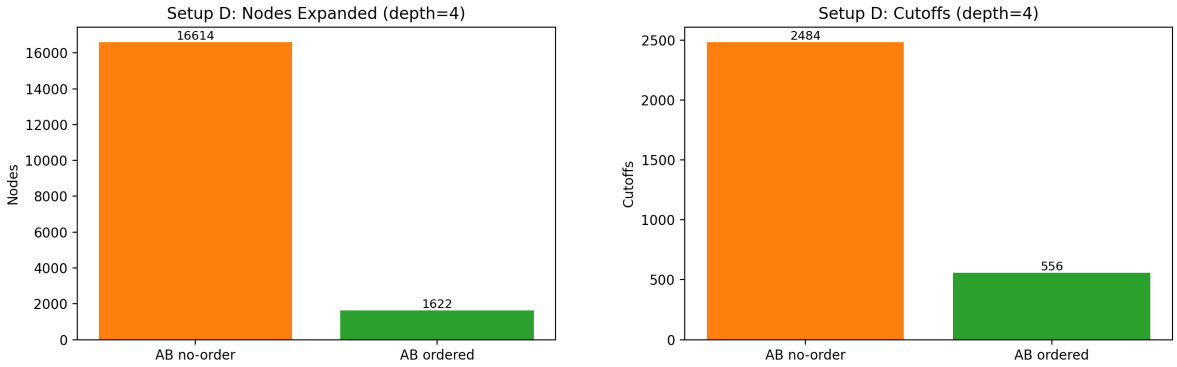
5.3 Setup C (fixed depth vs random): wins, nodes/move, time/move



(a) Win rate vs random (b) Avg. nodes per move (c) Avg. time per move

Figure 3: Setup C: Against random play, both search agents win reliably, while Alpha-Beta is significantly more efficient.

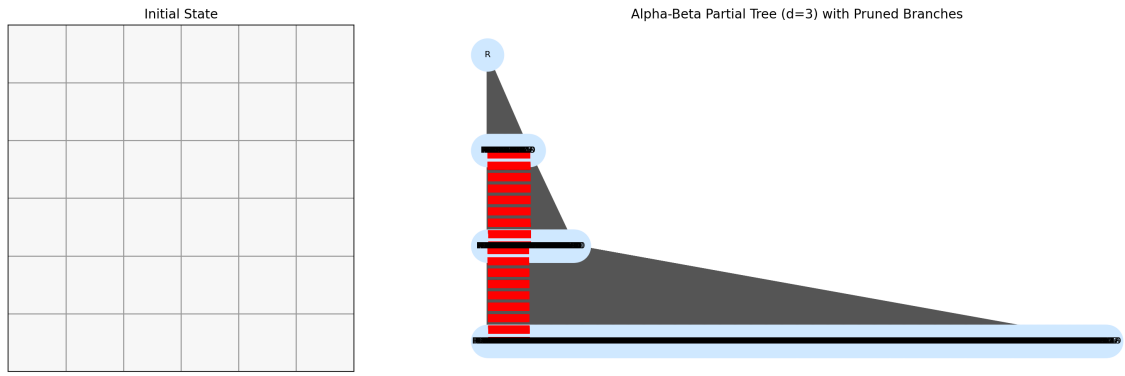
5.4 Setup D (move ordering impact): nodes, cutoffs, and pruned tree



(a) Nodes expanded

(b) Cutoffs

Figure 4: Setup D: Move ordering significantly increases Alpha-Beta pruning effectiveness.

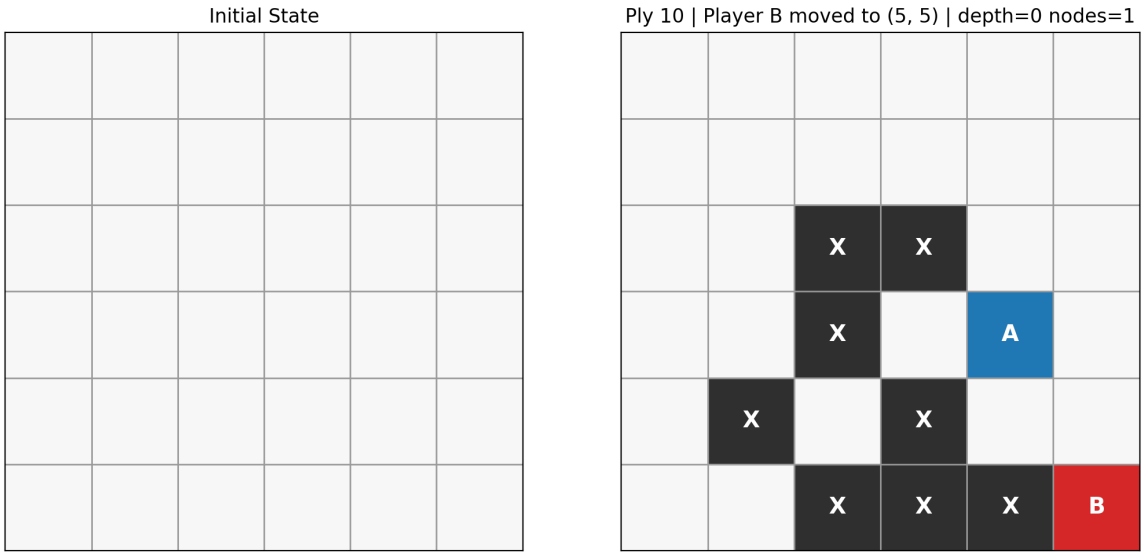


(a) Example start state

(b) Partial search tree with pruned branches

Figure 5: Visualization of Alpha-Beta pruning (ordered). Pruned branches are omitted/marked.

6 Qualitative Gameplay Snapshots



(a) Initial placement phase

(b) Later state after several plies

Figure 6: Two snapshots from Alpha-Beta vs Random. Blocked squares accumulate as pieces move.

7 Reflections and Observations

- **Efficiency vs strength:** At fixed depth, Alpha-Beta mainly improves efficiency (fewer nodes/time). Under time limits, the same efficiency often converts to **greater depth reached** (Fig. 2), which can improve playing strength against non-trivial opponents.
- **Move ordering matters:** Alpha-Beta pruning can range from modest to dramatic depending on move ordering, as shown by node and cutoff differences in Setup D.
- **Heuristic suitability:** Mobility difference is a natural fit for Isolation because the win condition is directly tied to restricting legal moves. It is simple, fast, and effective for depth-limited search.

8 Reproducibility

From the project root:

- Run tests: `python -m pytest -q`
- Run experiments (A–D): `python -m experiments.run_setups`
- Generate all figures + snapshots: `python -m visualization.auto_generate_figures`